

A
Learning Project-II Report
On

“NEST E-Commerce Website”

Submitted in partial fulfillment of
The requirements for the 4th Semester Sessional
Examination of

BACHELOR OF TECHNOLOGY
IN
COMPUTER SCIENCE & ENGINEERING

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CERTIFICATE

*This is to certify that the project work entitled “**NEST E-Commerce Website**” is done by **Rashmi Ranjan Sahoo, 23UG010660** in partial fulfillment of the requirements for the 4th Semester Sessional Examination of Bachelor of Technology in **Computer Science and Engineering** during the academic year 2024-25. This work is submitted to the department as a part of evaluation of 4th Semester Learning Project-II.*

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Rashmi Ranjan Sahoo (23UG010660)

ABSTRACT

The emergence of the internet has revolutionized the way we conduct business, and online shopping has become an integral part of modern commerce. This project focuses on the development shopping cart system, with HTML, CSS and JavaScript the primary goal is to create a seamless and user-friendly interface that enhances the overall online shopping experience.

The project begins with an exploration of the current landscape of e-commerce and the significance of an efficient scaling.

Shopping cart site with the needling reliance on digital transactions, it becomes imperative to design a platform that not only facilitates purchases but also ensures user satisfaction and engagement.

The front-end development phase involves the use of HTML to structure the content, CSS for styling, and JavaScript to add dynamic functionality The user interface is carefully crafted to provide an intuitive and visually appealing design. Responsive web design principles are implemented to ensure a consistent experience across various devices, catering to the diverse needs of users.

One key aspect of the project is the implementation of a secure and seamless authentication system. Users are provided with the ability to create accounts, log in securely, and manage their profiles. This not only enhances the user experience but also lays the foundation for personalized services, such as order history and recommendations based on past purchases.

The online shopping cart is designed to be dynamic and adaptive, allowing users to add, remove, or modify items in real-time. The interactive features are implemented using JavaScript to create a fluid a and responsive interface The shopping cart is more than just a virtual basket, it is a dynamic component that reflects the user's choices instantly.

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INTRODUCTION

The rapid evolution of the digital landscape has fundamentally transformed the way individuals interact with businesses and make purchasing decisions. At the forefront of this paradigm shift is the ubiquitous practice of online shopping. The convenience and accessibility afforded by e-commerce platforms have not only redefined consumer behavior but have also prompted a reevaluation of traditional retail models. As part of this transformative wave, this project endeavors to delve into the online developing a robust and intuitive online shopping cart system.

In the contemporary marketplace, an online shopping cart is more than just a transactional tool; it is the digital counterpart to the physical shopping experience. It serves as the virtual basket where users store their selections, make decisions, and ultimately proceed to checkout. Recognizing the pivotal role of this component, the project concentrates on the front-end development of the online shopping cart, employing HTML, CSS, and JavaScript to craft an engaging and dynamic user interface.

The significance of a well-designed online shopping cart cannot be overstated. It functions as the backbone of the entire e-commerce process, influencing user satisfaction, conversion rates, and the overall success of an online business. Therefore, this project places a premium on creating a seamless and user-friendly interface that not only facilitates transactions but also enhances the overall online shopping experience.

The initial stages of this project involve a comprehensive examination of the current state of e-commerce and the role played by online shopping carts in shaping user interactions. As technology continues to advance, consumer expectations rise in tandem. Users now demand not only efficiency in the transactional process but also a delightful and intuitive interface that mirrors the pleasures of physical shopping. This necessitates a nuanced approach to front-end development, incorporating the latest web technologies and responsive design principles.

HTML, as the foundational markup language, provides the structural framework for the online shopping cart. It delineates the various elements of the web page, laying the groundwork for a well-organized and logically structured interface. CSS steps in to add the aesthetic layer, defining the visual style and layout. The judicious use of CSS ensures a visually appealing design that aligns with contemporary design trends while prioritizing clarity and user engagement.

The dynamic functionality of the online shopping cart is brought to life by JavaScript, a versatile scripting language that enables real-time interactions. This project harnesses the power of JavaScript to create a fluid and responsive user interface. Users can effortlessly add or remove items from their virtual carts, and the interface dynamically updates to reflect these changes, providing instant feedback and enhancing the overall user experience.

Authentication and user management are integral components of the online shopping experience. Recognizing this, the project incorporates a secure and user-friendly authentication system. Users are empowered to create accounts, log in securely, and manage their profiles, laying the groundwork for a personalized and secure shopping journey. This not only enhances the user experience but also establishes a foundation for future features such as order tracking and personalized recommendations.

As we navigate through the intricacies of front-end development for the online shopping cart, it is crucial to acknowledge the evolving nature of user expectations. Accessibility and inclusivity are paramount considerations in the design process.

In the subsequent sections of this project, we delve deeper into the specific features and functionalities implemented to enhance the online shopping cart. From product categorization and search functionality to security protocols and potential future enhancements, each aspect is meticulously examined. This project contributes to the ongoing discourse on crafting engaging, secure, and accessible digital experiences in the ever-evolving realm of online shopping.

2.1 Purpose

The purpose of this project extends beyond the development of a conventional online shopping cart, it is an exploration and embodiment of a unique and engaging user experience within the vibrant, with its rich visual storytelling and diverse genres, has garnered a dedicated and passionate global fanbase.

The overarching goal is to merge the functionalities of a standard online shopping cart . By infusing the shopping cart interface with anime-inspired designs, graphics, and interactive features, the project endeavors to create an online shopping experience that resonates with the visual language and storytelling nuances of anime culture.

1) Convenience:- E-commerce websites allow customers to shop from anywhere at any time, providing a hassle-free shopping experience¹.

2) Cost Efficiency:- Operating an e-commerce site is generally cheaper than maintaining a physical store, reducing costs related to rent, utilities, and staffing².

3) 24/7 Availability:- E-commerce sites are always open, enabling businesses to make sales and provide customer support around the clock¹.

4) Customer Insights:- These websites can track customer behavior and preferences, helping businesses tailor their offerings and marketing strategies.

5) Scalability:- E-commerce platforms can easily scale to accommodate growing business needs without significant additional investment

2.2 Project Scope

The scope of an e-commerce website is vast, enabling businesses to reach a global audience, streamline sales, and offer 24/7 shopping. With the rise of online shopping and digital payments, e-commerce platforms can enhance customer experience, boost revenue, and provide data-driven insights, making them essential in today's digital economy.

1. Front-End Development:-

The primary focus of the project lies in the Dom-end development of the online shopping cart. This includes the utilization of IFTML, CSS, and JavaScript to craft a visually

appealing, responsive, and user-friendly interface. The front-end design will be centered around anime-inspired aesthetics, interactive themes, and interactive elements that resonate with the visual language of anime culture.

2. User Authentication and Personalization:-

The project involves the implementation of a secure and seamless user authentication system. Users can create accounts, log in securely, and personalize their profiles with anime-themed avatars or other visual elements. This personalization extends to the shopping experience, with the platform offering tailored recommendations based on the user's anime preferences, purchase history, and interaction patterns.

3. Product Catalog with Anime-Themed Categorization:-

The shopping cart will feature a diverse catalog of anime-related merchandise, ranging from standard items to exclusive and niche products. Products will be categorized thematically, allowing users to explore items based on their favorite anime series, genres, characters. The goal is to create a curated and thematic shopping experience that aligns with the diverse interests within the anime community

4. Community Engagement Features:-

To foster a sense of community among anime enthusiasts, the project includes features that encourage user interaction. This may involve the integration of user-generated reviews, discussion forums, and social media sharing functionalities. The platform aims to be a digital hub where anime fans can not only shop but also engage in discussions, share recommendations, and connect with like-minded individuals.

5. Secure Online Transactions:-

Security is a paramount consideration in online transactions. The project will implement protocols safeguard user data, including encryption for financial transaction and protection common web vulnerability. A secure and trustworthy platform is essential to in till confidence encouraging them to make online purchases with peace of mind.

6. Responsive Design for Various Devices:-

The scope extends to ensuring a consistent and enjoyable user experience across various devices. The end design will be responsive, adapting seamlessly to different screen sizes and resolutions. Whether users access the platform from a desktop, tablet, or mobile device, the interface will maintain its visual appeal and functionality.

7. Future Expansion and Innovation:-

While the initial scope focuses on creating a feature-rich anime-themed shopping cart, the project is designed with scalability in mind. It provides a foundation for future enhancements and innovations, such as the integration of augmented reality for product try-ons, interactive storytelling elements, and more features. The project aims to evolve in tandem with technological advancements and the changing expectations of the anime community.

In summary, the scope of the Shopping Cart project encompasses the development of a visually captivating, secure, and community-centric online shopping experience tailored specifically for anime enthusiasts. It seeks to go beyond the conventional boundaries of e-commerce, creating a digital space that celebrates the passion for anime while providing a seamless and personalized shopping journey.

2.1 PRODUCT FEATURE

The anime-themed Shopping Cart project incorporates a diverse range of features, meticulously designed to enhance the online shopping experience for anime enthusiasts. These features are geared towards creating an immersive, community-driven platform that goes beyond standard e-commerce. The following highlights the key features of the anime-themed shopping cart:

1. Inspired Front-End Design:-

- Visually appealing interface with anime-themed graphics, colors, and layout
- Responsive design for optimal viewing on various devices, including desktops, tablets, and mobile phones

- Engaging animations and transitions inspired by anime aesthetics

2. User Authentication and Personalization:-

- Secure user authentication system for account creation and login.
- Personalized user profiles with the ability to set anime-themed avatars
- Tailored shopping experience with recommendations based on user preferences

3. Thematic Product Catalog:-

- Diverse catalog of anime-related merchandise, including figures, apparel, accessories, and collectibles
- Thematic categorization based on anime series, genres, and characters
- Featured and exclusive sections for limited edition and niche products.

4. Community Engagement:-

- User-generated reviews and ratings for products.
- Discussion forums to facilitate community interaction and recommendations
- Social media integration for sharing favorite products and purchases.

5. Dynamic Shopping Cart Functionality:-

- Real-time updates to the shopping cart as users add or remove items
- Wishlist feature for users to save and track desired items.
- Seamless checkout process with multiple payment options

6. User-Friendly Interface:-

- Streamlined and intuitive design to ensure a seamless and user-friendly navigation experience
- Clear and concise product information with visually appealing layouts

- Guided onboarding process for new users to familiarize themselves with the platform

7. Trending Section and User-Generated Content:-

- Dedicated "Trending Now" section showcasing popular and in-demand products within the community

- User-generated content feed highlighting community favorites, reviews, and shared experiences

- Integration of trending tags and categories to assist users in discovering the latest and most talked-about anime merchandise.

8. Dynamic User Profiles:-

- Enhanced user profiles with dynamic activity feeds, showcasing recent purchases, reviews, and community interactions

- Option for users to share and display their anime-related interests, creating a sense of identity within the community

- Personalized profile badges or achievements based on user engagement and loyalty.

9. Future Expansion and Innovation:-

- Framework for continuous updates and enhancements based on user feedback.

- Exploration of emerging technologies for further innovation

- Consideration of features like interactive storytelling experiences or virtual events for a holistic anime themed platform

These features collectively contribute to the creation of a vibrant, secure shopping cart. By integrating elements inspired by modern culture and providing a platform for interaction and personalization, the project aims to redefine the online shopping experience for music dedicated.

WORK DONE IN RELATED AREA

In an e-commerce website like Anon, the work can include a wide range of activities across different functional areas. Here are five key tasks often involved:

1. Website Development:-

Building and maintaining the website's front-end (user interface) and back-end (server-side operations). This includes creating responsive designs, optimizing load times, and ensuring the site functions smoothly.

2. Product Management:-

Adding, updating, and categorizing products on the platform. This includes managing inventory, pricing, and product details to provide accurate information to customers.

3. User Experience (UX) and User Interface (UI) Design:-

Designing a user-friendly and visually appealing layout. This involves ensuring an intuitive shopping experience, easy navigation, and smooth checkout process.

4. Payment Gateway Integration :-

Setting up and managing secure payment options for customers, such as credit/debit cards, digital wallets, and other payment methods. This requires strong security to protect customer information.

5. Customer Service and Support :-

Providing customer support through various channels to address enquiry, handle returns and refund, and resolve issues, ensuring a positive customer experience.

SYSTEM ANALYSIS

Here's a system analysis outline for an anonymous e-commerce website, focusing on the main functional requirements, non-functional requirements, and system design. This would include steps such as defining user requirements, security needs, and analyzing how the system handles payments, product browsing, and anonymity.

1. System Overview

-> Objective:- To design an e-commerce website allowing users to browse, purchase, and review products anonymously.

-> Scope:- The system must include product listing, user accounts, order management, secure transactions, and anonymized user data handling.

2. Requirements Analysis

A) Functional Requirements:-

1. User Registration and Anonymity:- Users can register and log in without disclosing personally identifiable information (PII).

2. Product Browsing and Search:- Users can browse products by category, search by keyword, and filter results.

3. Shopping Cart Management:- Users can add/remove items to/from the cart, view total cost, and proceed to checkout.

4. Anonymous Checkout:- Allow purchases without requiring identifiable information; pseudonyms and secure payment methods like cryptocurrency might be used.

5. Order Tracking and Management:- Provide users with a way to track orders and view order history while preserving anonymity.

6. Product Reviews:- Users can post anonymous reviews and rate products without linking to their identity.

B) Non-Functional Requirements:-

- 1. Security:-** Implement strong encryption, secure transactions, and anonymized databases.
- 2. Performance:-** Efficient product searches, optimized loading times, and fast processing of transactions.
- 3. Scalability:-** Ability to handle an increasing number of users and products without compromising performance.
- 4. Reliability:-** Ensures uptime and consistency in order processing and product browsing.
- 5. Usability:-** A user-friendly interface that is easy to navigate while ensuring anonymity.
- 6. Privacy:-** Strict adherence to data privacy standards; only essential data should be stored and should be anonymized.

3) System Design

A) Architecture:-

Use a multi-tier architecture:-

Frontend:- A web application accessible on different devices (e.g., desktop, mobile). Technologies could include React or Angular for a responsive UI.

Payment Processing:- Integrate with an external payment provider that allows anonymous payments.

B) Data Storage and Anonymization:-

->Minimize data storage for user interactions. Avoid storing PII and instead use hashed or pseudonymous identifiers.

->Encrypt all transaction-related data to prevent exposure of user activity.

C) Security Measures:-

Encryption:- Apply SSL/TLS for data in transit, and encrypt sensitive data at rest.

Authentication and Authorization:- Enable multi-factor authentication (MFA) for additional account security.

Data Anonymization:- Implement hashing and tokenization techniques for data storage and usage.

D) Compliance:-

->Ensure compliance with privacy regulations such as GDPR by anonymizing data and obtaining consent for any data stored.

->Conduct regular security audits and vulnerability testing to safeguard user data.

4) Workflow Analysis

A) User Workflow:-

1. User visits the website and can browse products anonymously.
2. Adds items to their cart and proceeds to checkout with a pseudonym if desired.
3. Selects a payment method that supports anonymity.
4. Completes the purchase and receives a tracking ID to monitor the order.

B) Admin Workflow:-

1. Manage product listings and ensure data is anonymized.
2. Monitor transactions, conduct system audits, and verify that security protocols are maintained.
3. Review system performance metrics and resolve any issues affecting reliability or anonymity.

5) Potential Challenges and Solutions

Challenge-1: Maintaining user anonymity while handling transactions.

Solution: Use cryptocurrency or third-party services for anonymous payments, coupled with end-to-end encryption.

Challenge-2: Ensuring product reviews remain anonymous and authentic.

Solution: Develop an algorithm to detect spam reviews while keeping user identities hidden.

Challenge-3: Compliance with evolving privacy laws.

Solution: Regular audits and updates to privacy policies to stay current with regulations.

4.1 Hardware Requirements

The hardware requirements for the Anime-Themed Online Shopping Cart are formulated to provide a seamless, responsive, and visually captivating user experience for the diverse anime community. The following specifications outline the recommended hardware configuration for optimal performance:-

CLIENT-SIDE HARDWARE:-

The client-side hardware encompasses devices used by users to access the online shopping platform.

A) Desktops/Laptops:-

- Processor: Dual-core processor or higher.
- RAM: Sufficient for optimal performance.
- Storage: SSD recommended for faster data access.
- Graphics: Integrated or dedicated GPU for enhanced visual experiences.

B) Tablets:-

- Processor: Quad-core processor or higher.
- RAM: Adequate for responsive performance.
- Storage: SSD or eMMC storage for quick data access.
- Display: High-resolution display for an immersive visual experience.

C) Mobile Phones:-

- Processor: Quad-core processor or higher.
- RAM: Sufficient for smooth navigation.
- Storage: eMMC storage for efficient data access.
- Display: High-resolution, responsive touch screen.
- Sufficient storage capacity for the growing product catalog and user data.

4.2 Software Requirements

For analyzing the software requirements of an anonymous e-commerce website, it's crucial to address both functional and non-functional requirements. Here's an outline:

1. Functional Requirements:-

A) User Management:-

User Registration: Enable users to create accounts while keeping identities anonymous (use minimal data like email/username and possibly two-factor authentication).

User Authentication: Allow secure login/logout functionalities with privacy in mind.

Profile Management: Enable users to manage preferences, order history, and anonymous payment details.

B) Product Catalog Management:-

Product Listing: Display products with details (name, price, description, images) in categories.

Search and Filter: Allow users to search and filter products based on categories, price range, and other attributes.

C) Shopping Cart:-

Add to Cart: Allow users to add/remove products from their cart.**Cart Summary:** Display items in the cart, including total price and item details.

D) Order Management:-

Order Placement: Allow users to review and place orders anonymously.

Order Tracking: Provide anonymous order tracking with unique codes.

E) Payment Processing:-

Anonymous Payment Options: Offer payment methods that do not require personal details.

Order Confirmation: Send confirmation notifications without requiring personally identifiable information.

F) Notification System:-

Order Status Updates: Inform users about order statuses via anonymous methods (e.g., email or in-app messages).

2. Non-Functional Requirements:-

A) Security:-

Data Encryption: Use end-to-end encryption for data storage and transfer, especially for payment and order data.

User Privacy: Avoid collecting unnecessary data, using anonymous identifiers to protect identities.

Two-Factor Authentication: Implement optional two-factor authentication for increased security.

B) Performance:-

Scalability:- The system should handle a large number of users and products efficiently.

Response Time:- Aim for fast page load times and quick response during peak shopping times.

C) Reliability:-

Uptime: Ensure high availability, particularly during peak shopping periods.

Backup and Recovery: Implement a secure data backup and recovery system.

D) Usability:-

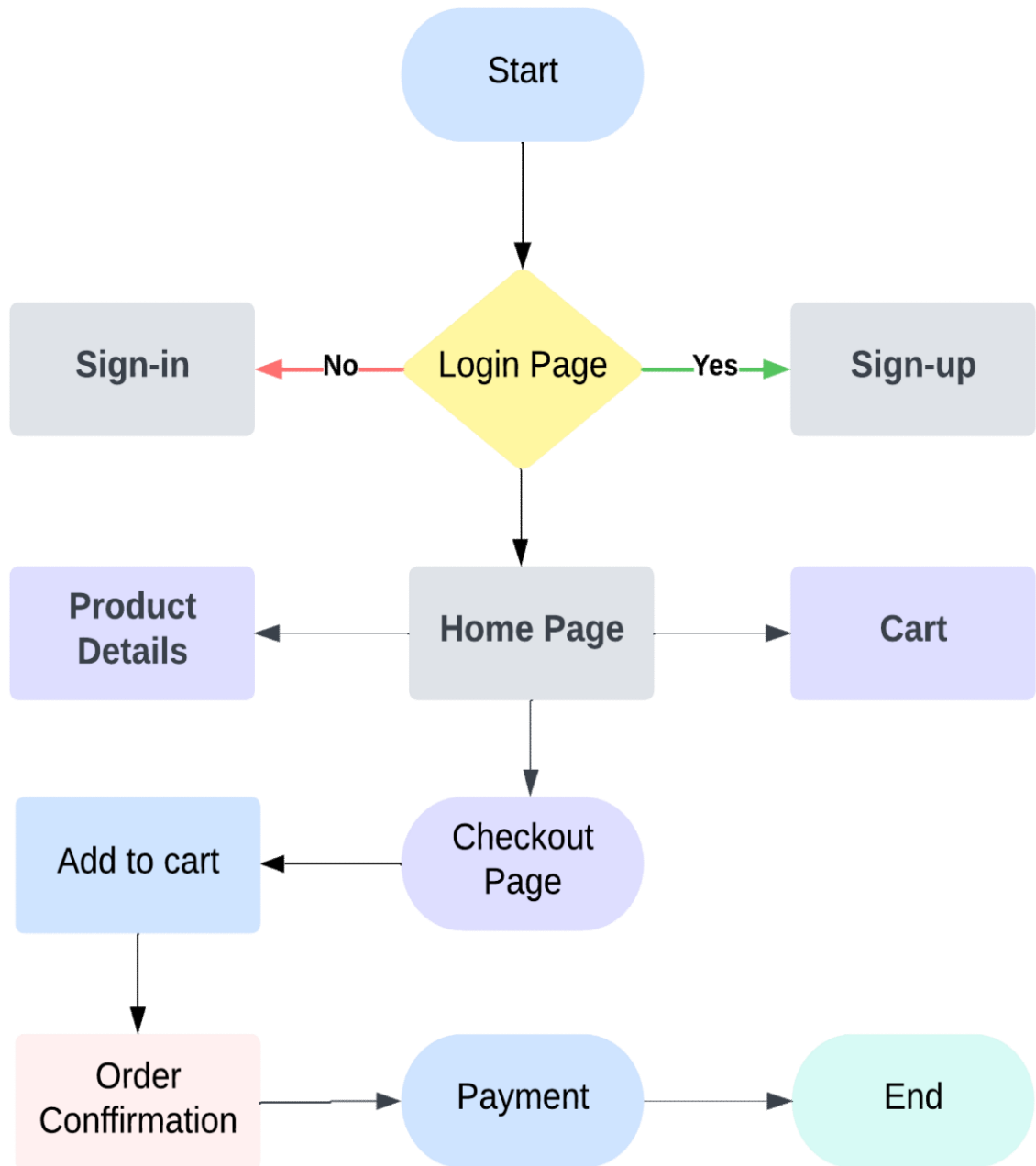
User Interface (UI): Design a simple, user-friendly interface focusing on ease of use and privacy.

Mobile Compatibility: Ensure the site is responsive and works on mobile devices.

SYSTEM DESIGN & SPECIFICATION

5.1 High Level Design (HLD)

FlowChart



5.2 Low Level Design (LLD)

5.2.1 Process Specification (Pseudo code / Algorithm)

A) User login

```
FUNCTION login_user(username, password):
    hash_password = hash(password)
    IF username exists in database AND hash_password matches:
        generate_session_token(username)
        RETURN "Login successful"
    ELSE:
        RETURN "Invalid username or password"
```

B) Browse product

```
FUNCTION browse_products(filter_criteria):
    products = fetch_products_from_database(filter_criteria)
    RETURN products
```

C) Product search

```
FUNCTION search_products(search_query):
    results = search_database(search_query)
    RETURN results
```

E) Add to cart

```
FUNCTION add_to_cart(product_id, quantity, session_token):
    cart = fetch_cart_from_session(session_token)
    IF product_id in cart:
        update_quantity(cart, product_id, quantity)
    ELSE:
        add_product_to_cart(cart, product_id, quantity)
    RETURN "Product added to cart"
```

D) View product detail

```
FUNCTION view_product(product_id):
    product_details = fetch_product_details(product_id)
    RETURN product_details
```

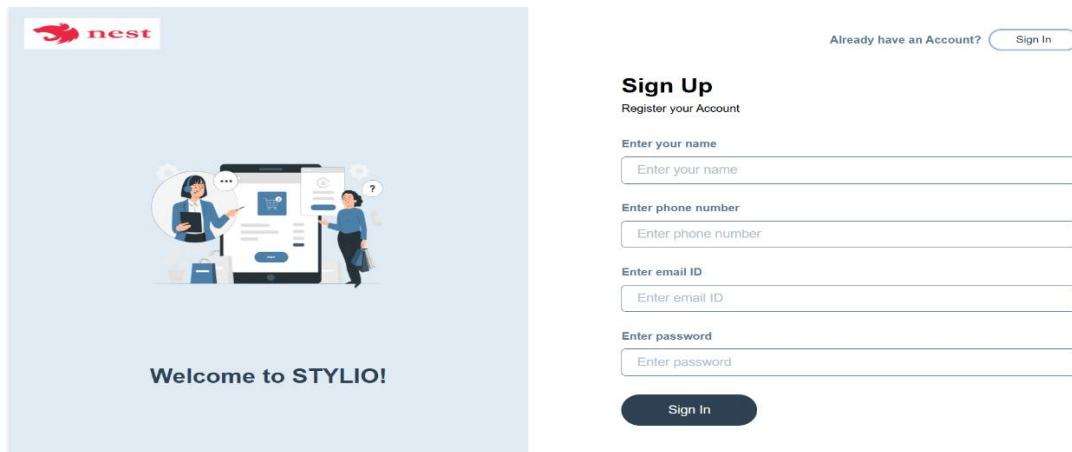
F) View cart

```
FUNCTION view_cart(session_token):
    cart = fetch_cart_from_session(session_token)
    RETURN cart
```

G) Check out process

```
FUNCTION checkout(session_token):
    cart = fetch_cart_from_session(session_token)
    IF cart is NOT empty:
        total_amount = calculate_total(cart)
        RETURN total_amount, "Proceed to payment"
    ELSE:
        RETURN "Cart is empty"
```

5.2.2 Screen-Shot Diagram



The screenshot shows the 'Sign Up' page for STYLIO. On the left, there is a light blue banner with the 'nest' logo at the top left. Below the logo is an illustration of two people interacting with a large smartphone. At the bottom of the banner, it says 'Welcome to STYLIO!'. On the right, there is a white background with a dark blue 'Sign In' button at the top right. Below this, the heading 'Sign Up' is followed by the subtext 'Register your Account'. There are four input fields: 'Enter your name', 'Enter phone number', 'Enter email ID', and 'Enter password'. Each field has a small icon to its left. At the bottom, there is a dark blue 'Sign In' button.

nest

Already have an Account? [Sign In](#)

Sign Up

Register your Account

Enter your name

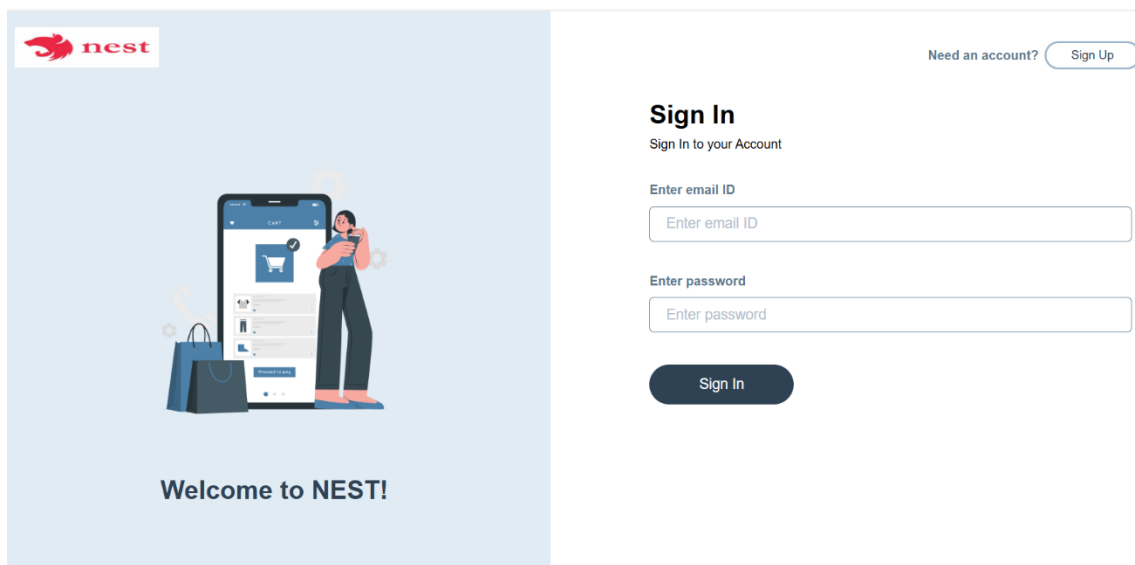
Enter phone number

Enter email ID

Enter password

[Sign In](#)

FIGURE 1



The screenshot shows the 'Sign In' page for NEST. On the left, there is a light blue banner with the 'nest' logo at the top left. Below the logo is an illustration of a person holding a large smartphone. At the bottom of the banner, it says 'Welcome to NEST!'. On the right, there is a white background with a dark blue 'Sign In' button at the top right. Below this, the heading 'Sign In' is followed by the subtext 'Sign In to your Account'. There are two input fields: 'Enter email ID' and 'Enter password'. Each field has a small icon to its left. At the bottom, there is a dark blue 'Sign In' button.

nest

Need an account? [Sign Up](#)

Sign In

Sign In to your Account

Enter email ID

Enter password

[Sign In](#)

FIGURE 2

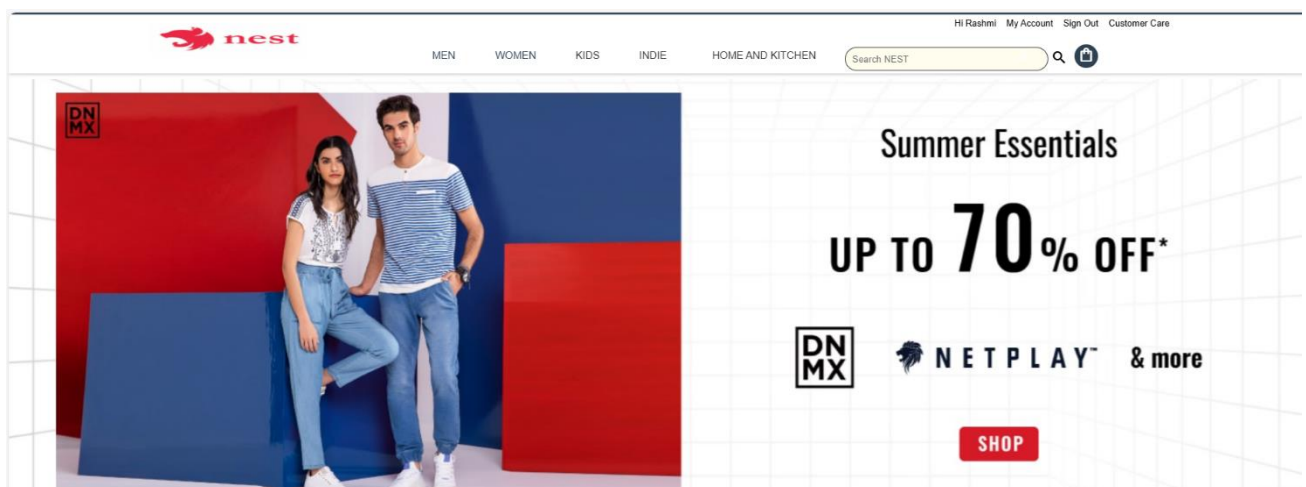


FIGURE 3

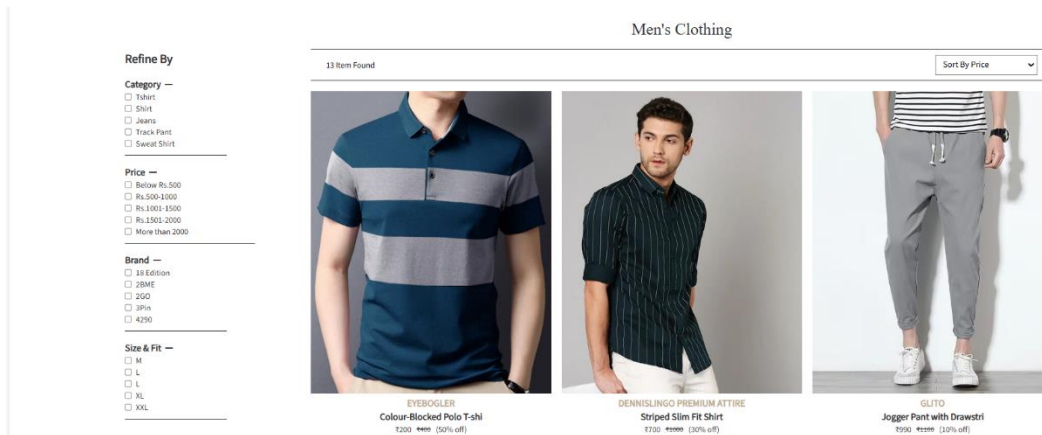


FIGURE 4

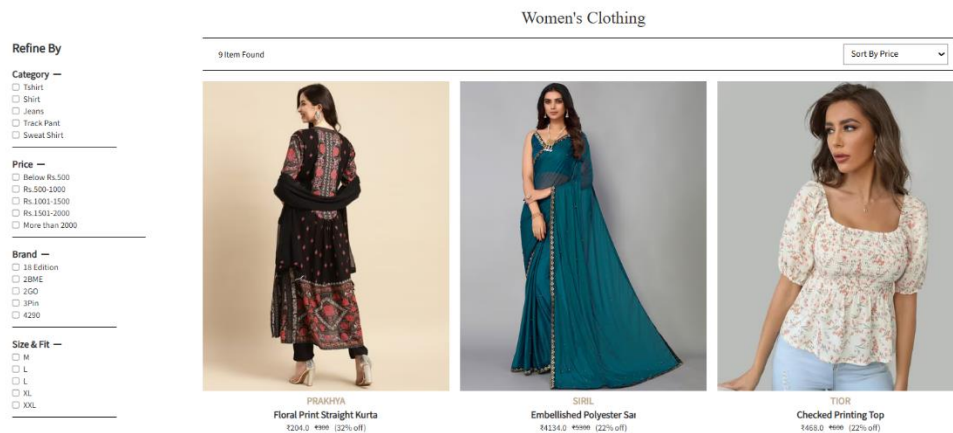


FIGURE 5

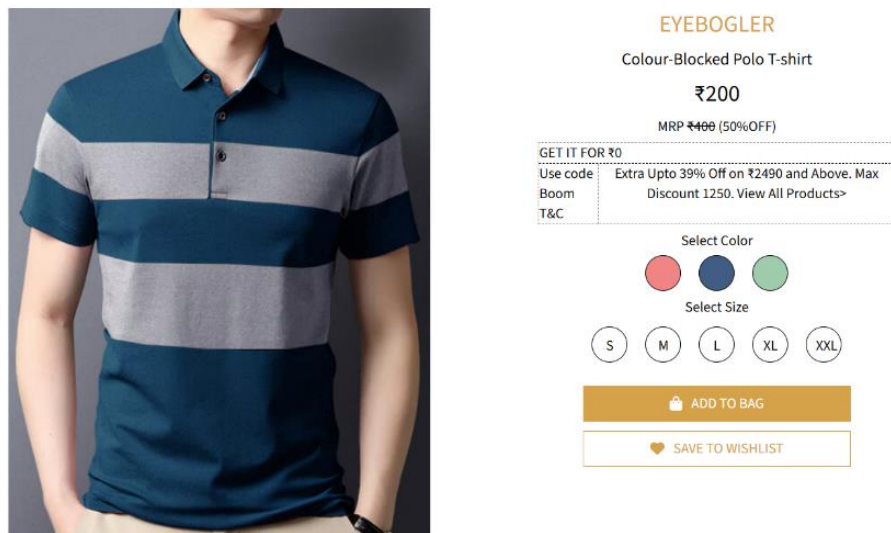


FIGURE 6

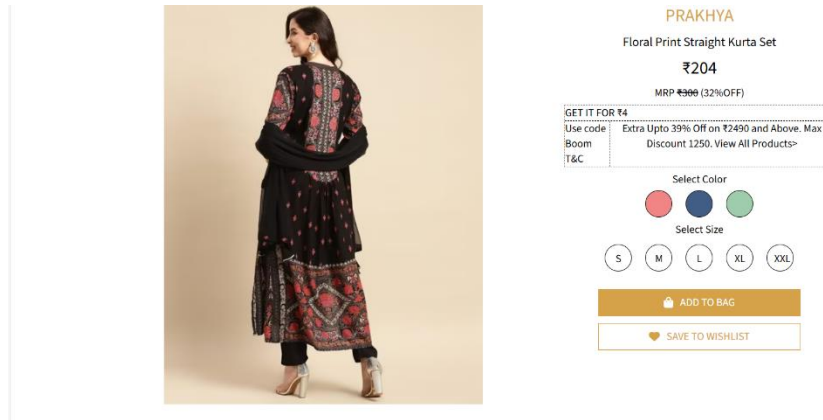


FIGURE 7

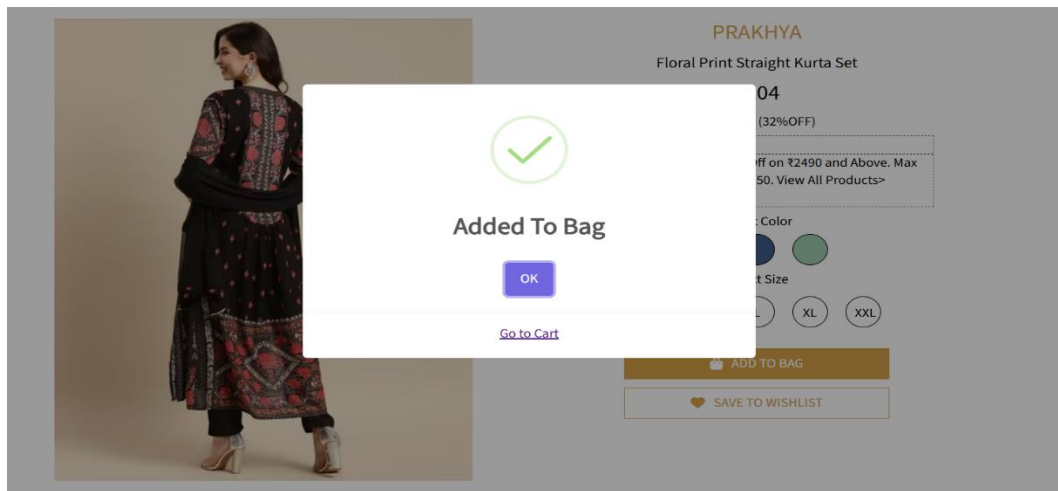


FIGURE 8

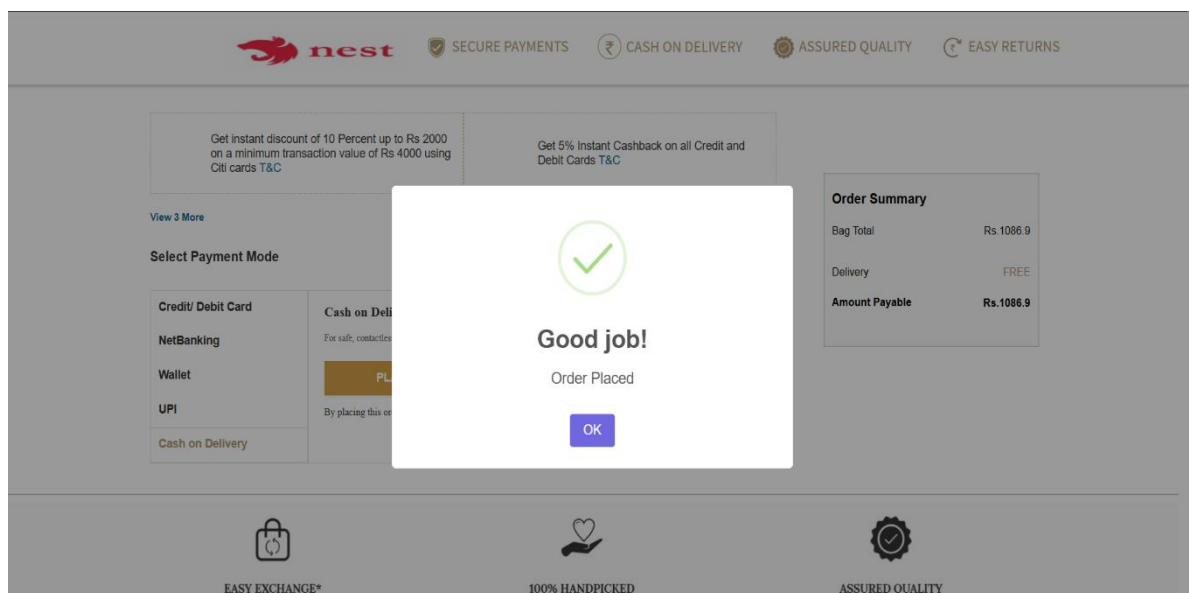


FIGURE 9

CODING

```
<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="UTF-8" />
    <meta http-equiv="X-UA-Compatible" content="IE=edge" />
    <meta name="viewport" content="width=device-width, initial-scale=1.0" />

    <link rel="stylesheet" href="./styles/index.css" />
    <link rel="stylesheet" href="./styles/swiper.css" />
    <link rel="stylesheet" href="./styles/footer.css" />
    <link
      rel="stylesheet"
      href="https://cdn.jsdelivr.net/npm/swiper@9/swiper-bundle.min.css"
    />
    <link
      href="https://fonts.googleapis.com/icon?family=Material+Icons+Outlined"
      rel="stylesheet"
    />
    <link rel="icon" href="./images/logo.jpg" />
    <title>
      Stylio.com | Shop Online the Latest Fashion in apparels brands in india
    </title>
  </head>

  <body>
    <div id="navbar">
      <div>
        <a href="./index.html">
</a>

</div>

<div class="one" id="men\_link">

<a href="/productsPage.html">

<p>MEN</p>

</a>

</div>

<div class="one" id="women\_link">

<a href="/women.html">

<p>WOMEN</p>

</a>

</div>

<div class="one" id="kids\_link">

<p>KIDS</p>

</div>

<div class="one" id="indie\_link">

<p>Electronics</p>

</div>

<div class="one" id="home\_link">

<p>HOME AND KITCHEN</p>

</div>

<div id="subnavbar">

<div id="navbar\_info">

<div>

<a href="/Coustomer Care.html">

Coustomer Care

</a>

```

</div>
<div id="navbar_signin_btn">
 Sign In / Join NEST
</div>
</div>
<div>
 <div>
 <h4>CLOTHING</h4>
 <h4>FOOTWEAR & ACCESSORIES</h4>
 <h4>ALL THATS NEW</h4>
 <p>Clothing</p>
 <p>Footwear</p>
 <p>Accessories</p>
 <h4>INTERNATIONAL BRANDS</h4>
 <h4>WINTERWEAR</h4>
 <h4>NIGHT & LOUNGEWEAR</h4>
 <h4>GROOMING</h4>
 <h4>ETHNIC & FESTIVE</h4>
 <p>Kurtas And Shirts</p>
 <p>Nehru Jackets</p>
 </div>
</div>

 <p>Footwear</p>
 <p>Accessories</p>
 <h4>INTERNATIONAL BRANDS</h4>
 <h4>WINTERWEAR</h4>
 <h4>NIGHT & LOUNGEWEAR</h4>
 <h4>GROOMING</h4>
 <h4>ETHNIC & FESTIVE</h4>
 <p>Kurtas And Shirts</p>

```

```

 <p>Nehru Jackets</p>
 </div>
</div>
 <p>Footwear</p>
 <p>Accessories</p>
 <h4>INTERNATIONAL BRANDS</h4>
 <h4>WINTERWEAR</h4>
 <h4>NIGHT & LOUNGEWEAR</h4>

<div class="horizontal-banner">

</div>

<div class="swiper mySwiper5">
 <div class="swiper-wrapper">
 <div class="swiper-slide">

<div class="swiper mySwiper7">
 <div class="swiper-slide">

 </div>
</div>

```

```

</div>
<div class="swiper-pagination"></div>
</div>

<div class="horizontal-banner">

</div>

<div class="swiper mySwiper8">
 <div class="swiper-wrapper">
 <div class="swiper-slide">

 </div>
 <div class="swiper-slide">

 </div>
 <div class="swiper-slide">

</div>
<div class="swiper-slide">

</div>
<div class="swiper-slide">

</div>
<div class="swiper-slide">

</div>
<div class="swiper-slide">

</div>
```

```

 </div>
 <div class="swiper-pagination"></div>
</div>

*{
 margin: 0;
 font-family: 'Zen Kaku Gothic New', sans-serif;
}
#logo {
 cursor: pointer;
}
#navbar {
 position: fixed;
 width: 100%;
 top: 0px;
 background-color: #FFFFFF;
 height: 85px;
 border-top: 3px solid #2F4254 ;
 display: flex;
 box-shadow: 0px 0px 5px #9a9a9a;
 z-index: 10;
}
#navbar > div:nth-child(1) {
 flex-basis: 20%;
 margin-left: 10.5%;
 padding-left: 5px;
 padding-top: 10px;
}
#navbar > div:nth-child(1) img {
 width: 230px;
}

```



```

#navbar > div:nth-child(n+2):nth-child(-n+6) {
 padding-top: 50px;
 cursor: pointer;
 box-sizing: border-box;
 text-align: center;
 color: #333;
 transition: 0.2s;
 font-size: 15px;
 font-weight: 400;
}
#navbar > div:nth-child(n+2):nth-child(-n+6):hover {
 border-bottom: 4px solid #2F4254;
 font-weight: bold;
}
#navbar > div:nth-child(2) {
 flex-basis: 5%;
}
#navbar > div:nth-child(3) {
 flex-basis: 6%;
}
#navbar > div:nth-child(4) {
 flex-basis: 5%;
}
#navbar > div:nth-child(5) {

```

```

let TotalCustomer=document.querySelector(".data-2");
let product=document.getElementById("product");
let pagination=document.querySelector("#pagination");

```

```

window.addEventListener("load",()=>{

```

```

 fetchrender("1");
 })

function fetchrender(page){
 fetch(`https://stylio.onrender.com/products?_limit=8&_page=${page}`, {
 method: "GET",
 headers: {
 "Content-Type": "application/json",
 }
 }).then((res) => {
 // console.log(`X-Total-Count`);
 let count = res.headers.get("X-Total-Count");
 Totalproduct.innerText = count;
 let button = Math.ceil(count/8);
 // console.log(button);
 pagination.innerHTML = null;
 for(let i=1; i<=button; i++){
 pagination.append(allbutton(i));
 }
 // console.log(count);
 return res.json();
 }).then((data) => {
 // console.log(data);

 product.innerHTML = null;
 let productdisplay = getall(data);

 }).catch((err) => {
 console.log(err);
 })
}

```

```

 })

}

function allbutton(i){
 let buttton=document.createElement("button");
 buttton.innerText=i;
 buttton.addEventListener("click",()=>{
 // console.log(i);
 fetchrender(i);
 })
 return buttton;
}

function getall(data){
 let div=document.createElement("div");
 div.classList.add("Allproduct");
 product.append(div);

 data.forEach((item)=>{
 let getproduct= getdata(item.title,item.price,item.img);
 div.append(getproduct);
 })

 return div;

}

function getdata(title,price,img){
 let div=document.createElement("div");

```

```
let Name=document.createElement("h5");
Name.innerText=title;
Name.style.color="#b19975";
```

```
let Price=document.createElement("p");
Price.innerText=`₹ ${price}`;
```

```
let Image=document.createElement("img");
Image.src=img;
```

```
div.append(Image,Name,Price);
```

```
return div;
}
```

```
// fetch order
let Totalorder=document.querySelector(".data-3")
fetch(`https://stylio.onrender.com/orders`)
.then((res)=>{
 return res.json();
}).then((data)=>{
 // console.log(data);
 // console.log(data.length);
 Totalorder.innerText=data.length;
 console.lo

}).catch((err)=>{
 console.log(err);
})
.catch((err)=>{
 console.log(err);
```

```

 })

 // fetch userdetails
 fetch(`https://stylio.onrender.com/userDetails`)
 .then((res)=>{
 return res.json();
 }).then((data)=>{
 // console.log(data);

 function change_info() {
 let info = document.getElementById("navbar_info");
 info.innerHTML = null;

 let div1 = document.createElement("div");
 let p1 = document.createElement("p");
 p1.id = "user_name";
 p1.innerText = "Hi " + name1;
 div1.append(p1);

 let div1 = document.createElement("div");
 let p1 = document.createElement("p");
 p1.id = "user_name";
 p1.innerText = "Hi " + name1;
 div1.append(p1);
 console.lo

 }).catch((err)=>{
 console.log(err);
 })
 .catch((err)=>{
 console.log(err);
 })
 }

```

```
})
```

```
// fetch userdetails
```

```
fetch(`https://stylio.onrender.com/userDetails`)
```

```
.then((res)=>{
```

```
 return res.json();
```

```
}).then((data)=>{
```

```
 // console.log(data);
```

```
function change_info() {
```

```
 let info = document.getElementById("navbar_info");
```

```
 info.innerHTML = null;
```

```
 let div1 = document.createElement("div");
```

```
 let p1 = document.createElement("p");
```

```
 p1.id = "user_name";
```

```
 p1.innerText = "Hi " + name1;
```

```
 div1.append(p1);
```

```
 let div1 = document.createElement("div");
```

```
 let p1 = document.createElement("p");
```

```
 p1.id = "user_name";
```

```
 p1.innerText = "Hi " + name1;
```

```
 div1.append(p1);
```

```
 let div1 = document.createElement("div");
```

```
 let p1 = document.createElement("p");
```

```
 p1.id = "user_name";
```

```
 p1.innerText = "Hi " + name1;
```

```
 div1.append(p1);
```

```

let div2 = document.createElement("div");
let p2 = document.createElement("p");
p2.innerText = "My Account";
p2.id = "my_account";
div2.append(p2);

let div3 = document.createElement("div");
let p3 = document.createElement("p");
p3.innerText = "Sign Out";
p3.id = "sign_out_btn";
div3.append(p3);

let div4 = document.createElement("div");
let p4 = document.createElement("p");
p4.innerText = "Customer Care";
div4.append(p4);

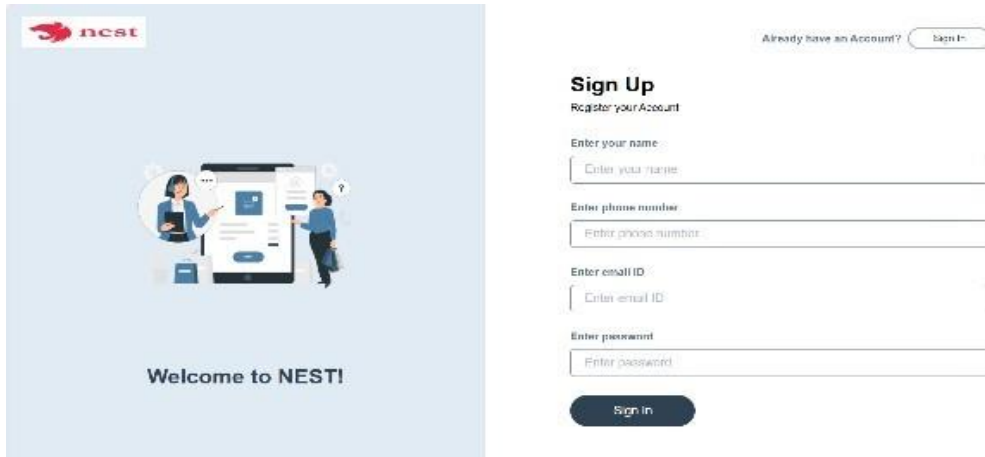
info.append(div4, div3, div2, div1);
}

let div3 = document.createElement("div");
let p3 = document.createElement("p");
p3.innerText = "Sign Out";
p3.id = "sign_out_btn";
div3.append(p3);

let div3 = document.createElement("div");
let p3 = document.createElement("p");
p3.innerText = "Sign Out";
p3.id = "sign_out_btn";
div3.append(p3);

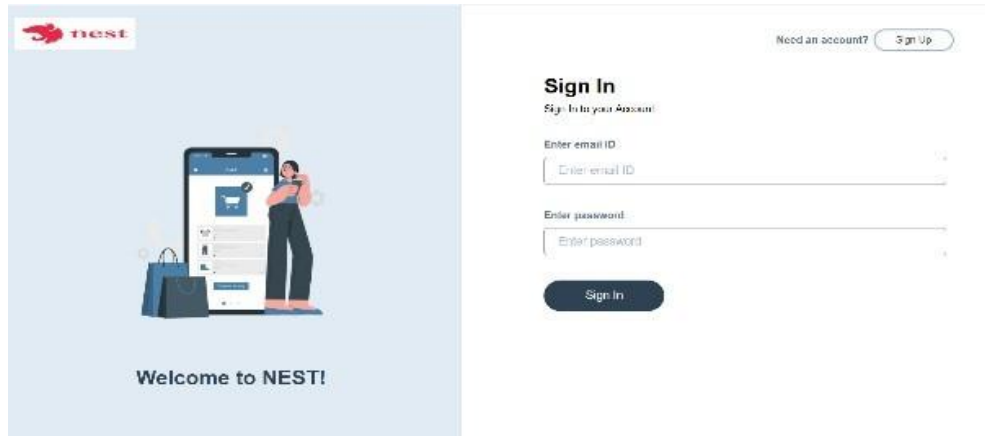
```

## TESTING



The image shows a 'Sign Up' form for NEST. On the left is a light blue panel with the NEST logo at the top, an illustration of a woman and a man interacting with a large smartphone, and the text 'Welcome to NEST!'. The right panel is white and contains the following elements: a link 'Already have an Account? Sign In' in a rounded rectangle; the heading 'Sign Up' with the subtext 'Register your Account'; four input fields labeled 'Enter your name', 'Enter phone number', 'Enter email ID', and 'Enter password'; and a dark blue 'Sign In' button at the bottom.

FIGURE 1



The image shows a 'Sign In' form for NEST. On the left is a light blue panel with the NEST logo at the top, an illustration of a man holding a smartphone, and the text 'Welcome to NEST!'. The right panel is white and contains the following elements: a link 'Need an account? Sign Up' in a rounded rectangle; the heading 'Sign In' with the subtext 'Sign In to your Account'; two input fields labeled 'Enter email ID' and 'Enter password'; and a dark blue 'Sign In' button at the bottom.

FIGURE 2

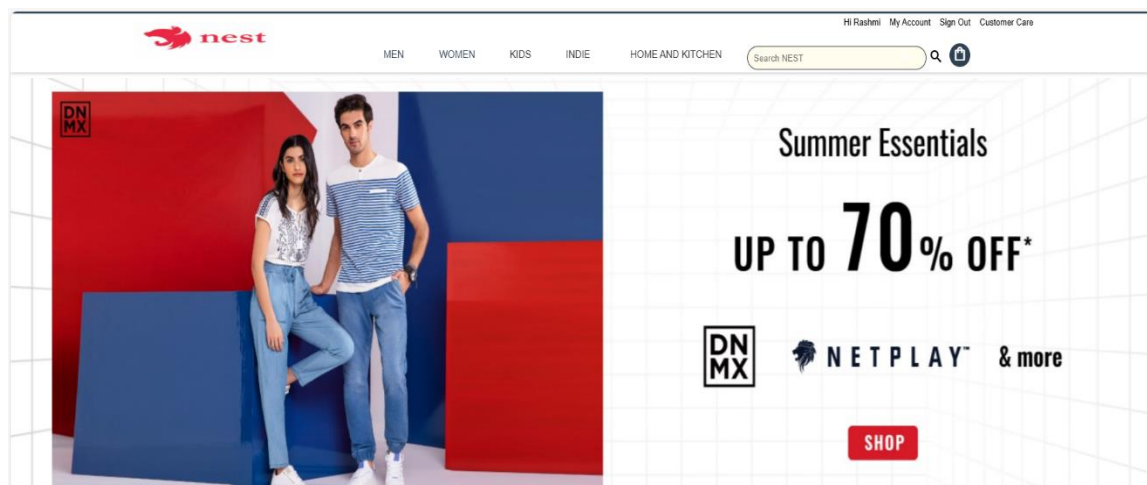


FIGURE 3



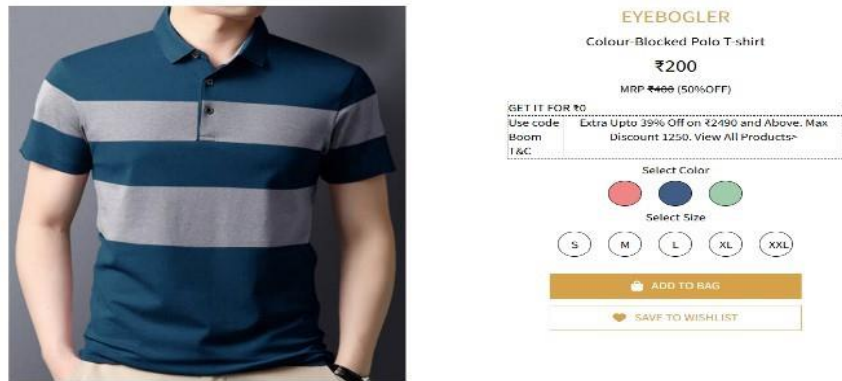


FIGURE 4

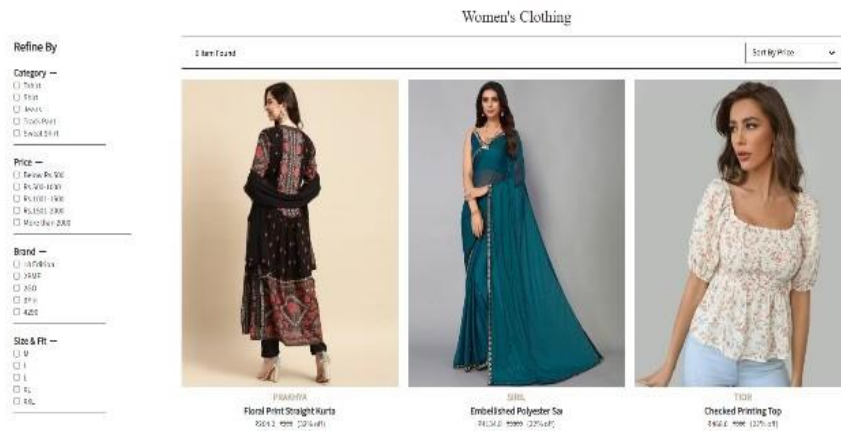


FIGURE 5



FIGURE 6

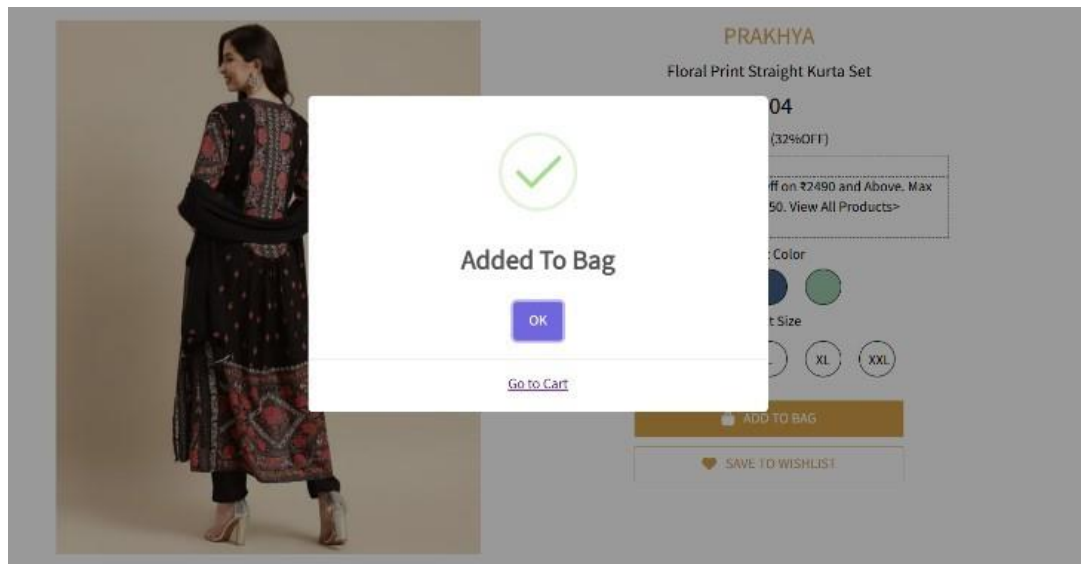


FIGURE 7

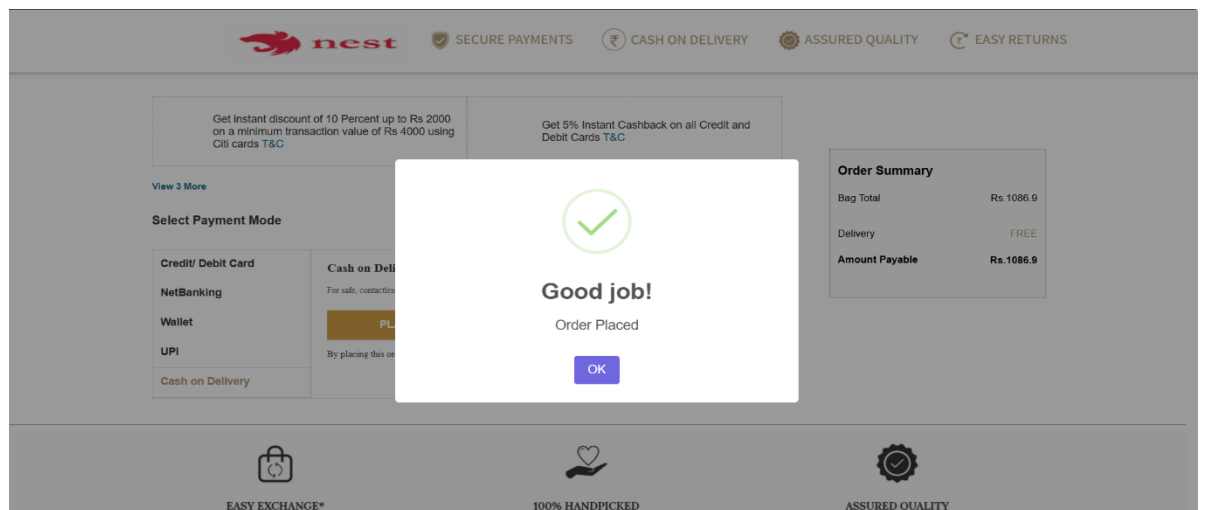


FIGURE 8

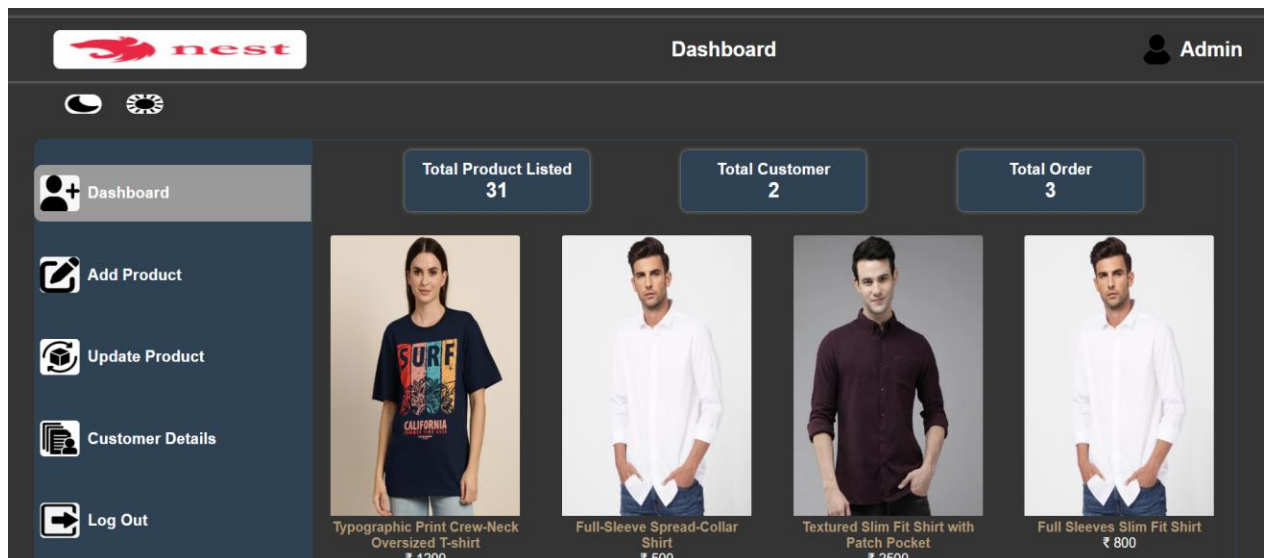


FIGURE 9

## **CONCLUSION & LIMITATION**

### **CONCLUSION :-**

In conclusion, the development of the e-commerce platform exemplifies a significant step forward in creating a seamless, user-centric, and secure online shopping experience. The project integrates essential features such as an intuitive user interface, robust security protocols, and responsive design to cater to diverse user needs, ensuring ease of navigation and user satisfaction. By incorporating advanced functionalities such as anonymous browsing, user authentication, and efficient product management, the platform addresses both practical e-commerce requirements and user privacy expectations.

The analysis of functional and non-functional requirements, alongside the implementation of multi-tier architecture, demonstrates the commitment to maintaining high performance, reliability, and scalability. These components collectively support a dynamic shopping environment that meets modern digital commerce standards. The inclusion of encrypted transactions and data anonymization further ensures user trust, aligning the platform with the best practices for privacy and security compliance.

## **LIMITATION :-**

While the development of this e-commerce platform has been comprehensive and strategic, there are certain limitations to acknowledge:

### **1) Data Privacy and Anonymity Constraints**

Although extensive efforts were made to ensure user anonymity and secure transactions, the use of third-party payment systems for anonymous payments may still pose potential data exposure risks. Ensuring complete user anonymity while complying with evolving data protection laws may require continuous adaptation and enhancement of privacy protocols.

### **2) Scalability Challenges:**

While the system has been designed with scalability in mind, rapid growth in user numbers or product catalogs could place pressure on server resources and response times. Additional investments in cloud infrastructure or load balancing solutions may be required to maintain optimal performance during peak usage periods.

### **3) Dependence on Internet Connectivity**

The platform's functionality is heavily dependent on stable internet connections. Users in regions with limited or unreliable internet access may face challenges in experiencing the platform's full capabilities, impacting user reach and overall satisfaction.

## **REFERENCES**

### **1) Mentors and Advisors**

-Special thanks to the project mentors who provided expert guidance throughout each stage, ensuring that best practices in design, development, and security were adhered to.

-Appreciation for their feedback on critical aspects of system architecture and user experience.

### **2) Teachers and Academic Instructors**

-Grateful acknowledgment to the professors and instructors who shared their deep knowledge in computer science and e-commerce principles, enriching the foundation needed for this project.

-Their lectures and resources were instrumental in shaping the approach taken to solve technical challenges.

### **3) Friends and Peers**

-Sincere thanks to friends who participated in brainstorming sessions, offering creative ideas and perspectives that inspired new features and improvements.

-Their informal testing and constructive feedback helped refine the platform and enhance user-centric elements.

### **4) Online Resources and Documentation**

Reference to relevant technical documentation, tutorials, and forums that provided solutions and clarifications for specific development challenges.

